

Matilde Dolfato

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EDUCATION

Bocconi University, Milan, Italy

M.Sc. Artificial Intelligence

Sep 2024–Present

- Courses: Linear Algebra, Algorithms for Opt and Inference, Deep Learning, Geometric Computer Vision, Information Theory, PDE, Complex Systems, Optimal Transport, Real and Fourier Analysis (audit)
- GPA: 29.2/30

B.Sc. Economic and Social Sciences

2021–2024

- Advisor: Prof. Jérôme Adda
- Final work: *Can Machine Learning Aid Econometrics?*, grade: 110/110 cum laude

Universitat Pompeu Fabra, Barcelona, Spain

Exchange Program

2024

Università degli Studi di Milano Statale

B.Sc. Biotechnology, Single Courses

2022–2023

EXPERIENCE

Lean Seminar organizer

Mar 2026–Present

Department of Decision Sciences, Bocconi

- Organizing and leading a *reading seminar* on the Lean proof assistant

Research fellow

Nov 2025–Dic 2025

NOVA Information Management School, Lisbon

- Supervisor: Prof. Mauro Castelli
- Research area: explaining *double descent* phenomenon, what underlies stochastic gradient descent preference for smoothness, what is smoothness

Visiting Student

Apr 2025–Nov 2025

Bocconi Institute for Data Science and Analytics (BIDSA)

- Mentor: Prof. Elia Brué
- Research area: Ricci curvature bounds for discrete spaces, optimal transport, convergence of diffusion processes and functional inequalities in spectral theory
- *Technical notes* and two presentations (1,2)

Founding and Board Member

Sep 2025–Present

Bocconi Student Association for Mathematical Sciences

- Gave a *lecture on optimal transport*

Reading Seminars Information Geometry (*Technical notes*)

Spring 2025

Research Internship Algebraic Topology

Spring 2025

Department of Decision Sciences, Bocconi

Research Assistant

Jun 2023–Dec 2023

Innocenzo Gasparini Institute for Economic Research, Bocconi

- Mentor: Prof. Massimo Marinacci
- Research area: re-interpretation and adaptation of the first quantitative analysis in social sciences by Cesare Beccaria (with tools of probability and decision theory)

SELECTED PROJECTS

saLM-1: Self-Aware Language Model Ensemble (LLM ensemble architecture)

Adversarial Robustness Pipeline (CNN adversarial training)

Wordle Guesser (Entropy-based guesser)

Movie Recommendation System (Web app)

SKILLS

Programming: Lean, Python (proficient), R, STATA (advanced), C, C++ (beginner)

Languages: Italian (native), English (proficient), Spanish (advanced)

OTHER INTERESTS

Sailing, surfing, windsurfing; volleyball (ex-agonistica); mountain hiking and skiing; piano; theater and cinema